

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 1

Data processing and information

87 questions from 31 question papers, Feb/March 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

9626/12 Feb/March 2022 · Question 1

[Mark scheme](#)

1 A collection of data could be the following:

1989, Paolo, $\sqrt{}$, ₹, $\neq$, OR

(a) Explain, by defining the term ‘data’, why this is just a collection of data and is not information.

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(b) Select **three** of these items of data. Explain what information each item might represent and how it could be used.

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- 6 A school secretary is updating a database by copying data from some hard copy records. Here are three of the records:

First_name	Family_name	Student_number	Previous_school	Date_of_birth
Jonathon	Odoki	12036	St Mary's	07/12/2004
Sven	Johansson	09132	The Grange	13/10/2007
Manjit	Sangherra	10102	Bluestars	11/03/2007

The Student_number will be stored as text, with the first two characters representing the year group the student belongs to.

The Student_number and Date_of_birth are the only data that need to be validated as they are entered. A range check, limit check or presence check would not be appropriate for validating these data.

- (a) Describe, in precise detail, **three** appropriate validation checks that could be used to validate the data. Each validation check must be different.

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(b) As well as validation, verification will need to be carried out.

Explain why verification is needed in addition to validation. Use examples from your answers to part (a).

[6]

1 The processing of customer orders is an example of batch processing.

Describe, using customer orders as an example, what is meant by batch processing.

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- 4 A supermarket uses a computer to automatically update stock levels. Each record in the **Product file** contains the fields shown.

Product file

Bar_code	Product	Number_in_stock	Reorder_level	Reorder_quantity	Supplier_ID
5022476130067	Oaters	131	130	250	B671841
0100108097858	Tea	182	130	310	A296593
0201570244453	Beans	317	212	450	C562148

When the Number_in_stock of a product item falls to the Reorder_level, that product must be reordered. The Reorder_quantity is the number of items of the product that must be reordered.

The **Supplier file** contains the contact details of all the suppliers.

Supplier file

Supplier_ID	Supplier	Contact
B671841	J Sullivan	js@cmail.co.ke
A296593	Bluants	bluants@hmail.ug
C562148	Golan	golan@smail.eg

Only some of the records in the Product and Supplier files are shown.

Describe, using the Oaters record as an example, the steps the computer would follow to update the Product file after a point of sale terminal reads the bar code for a packet of Oaters.

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9626/12 May/June 2022 · Question 2

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- 3** Julia is setting up a hairdressing business. She has recruited several staff. She is creating a database of all her staff and has written down all their details ready for entering into a computer. She will need to verify the data when it is entered.

She has been told that she cannot use a parity check but should use visual verification and double data entry.

- (a)** By describing what is meant by a parity check, explain why it might not be suitable in this scenario.

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- (b)** Compare the similarities and differences between the visual checking and double data entry methods of verification.

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Every week the master file has to be updated to include the new wage earned so far this year.

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(b) Updating a master file in a payroll system is an example of batch processing.

Discuss the advantages and disadvantages of using batch processing, compared to other types of processing, for a payroll system.

[8]

1 Paulo is carrying out a study on the level of noise caused by traffic in the streets around his home. He will need data for his study.

(a) Describe what is meant by direct data.

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(b) Describe how Paulo could use direct data sources for his study.

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- 3 Most companies use transaction files to update master files in their payroll systems. One part of the updating process is to calculate weekly or monthly pay. Another part is to update workers' personal and work details.

Without referring to the calculation of pay, describe the three types of transaction that would take place when workers' details are being updated. You must give an example for each type of transaction.

Transaction type 1:

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Example 1:

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Transaction type 2:

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Example 2:

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Transaction type 3:

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Example 3:

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Mark scheme

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1 Here is a set of data.

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Explain why this is just a set of data and describe how this particular set of data would become information.

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- 5 The owner of a book shop has created a database of her books. She will now test that it works. Some of the records are shown. The data shown contains the maximum and minimum values. There are only two types of book.

ISBN	Title	Author	Book_type	Copies
9781787632196	Blue Moon	Lee Child	H	4
9781858683416	Prelude to Foundation	Isaac Asimov	P	5
9780593072493	Inferno	Dan Brown	H	3
9781787461352	Revenge	James Patterson	P	2
9781529027488	One Good Deed	David Baldacci	H	5

- (a) Describe, in detail, the **three** most appropriate validation checks, apart from a check digit, type check or presence check, that could be used to validate the data. For each validation check, your description should include the field it could be used for. Each validation check and each field must be different.

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(b) The owner will also need to verify the data in her database.

Using examples from the three fields you used in part **(a)**, explain why verification is needed in addition to validation.

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8 Encryption is often used to protect data.

(a) Discuss the need for encryption.

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(b) Discuss the advantages and disadvantages of using symmetric encryption.

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Online processing can be used when goods are purchased.

(c) Define what is meant by online processing.

[1]

(d) Johann has previously registered with an online store.

Describe the steps he takes when purchasing goods from the online store.

[5]

5 Explain why encryption is needed.

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1 Encryption can be used to protect data.

(a) Define encryption.

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(b) Explain how intercepted data could be misused if it was not encrypted.

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- 5 D'Oliveira Inc. is a company which sells stationery products to many stores. The company uses a transaction file to keep a record of the number of products it has sold each month.

Part of a transaction file is shown. It contains:

- ProductID
- NumberSold (the number of that product sold in a particular month)

Part of the master file used by the company is also shown. It contains:

- ProductID
- Product (description of product)
- NumberinStock (how many of that product are left in stock)

The NumberinStock on the new master file is calculated every month by subtracting the NumberSold from the NumberinStock on the old master file.

Transaction file		Master File		
ProductID	NumberSold	ProductID	Product	NumberinStock
2210	6500	1965	Folder wallets	9000
3356	7300	2210	Ring binders	12000
4268	2900	2976	Notepads	10100
5690	4600	3356	A4 paper	13450
		4268	A3 paper	8800
		5690	HB pencils	9200
		6528	Pencil cases	11150
		7649	Maths sets	8240

Fig. 5.1

- (a) Identify the **two** processes that would have been performed on the transaction file before the master file could be updated.

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- (b)** List the steps involved when the master file is updated using the transaction file. Use the data shown in Fig. 5.1. Stop at the step where the third master file record is read.

[8]

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3 Many credit card purchases use online processing, whereas customer bills tend to be produced using batch processing.

Describe the differences between online processing and batch processing systems.

[5]

9626/13 May/June 2023 · Question 1

1 Describe what is involved in data logging.

[4]

2 Explain the purpose of the SSL/TLS protocol.

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- 3 There are different types of processing including batch processing and real-time processing.
- (a) Without referring to master files and transaction files, describe what is meant by batch processing.

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- (b) Rocket guidance systems are used in space exploration.
- Explain why a rocket guidance system is an example of the use of real-time processing.

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2 Two methods of checking the accuracy of data are validation and verification.

Describe the differences between validation and verification.

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4 Explain how HTTPS websites provide security of data.

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- 5 Maria wants to use information about the families in her town for a study. She will get the information from an indirect data source, the town’s electoral register. This is kept at the local council office. She has decided **not** to use a questionnaire, which is a direct data source, to gather the information she needs.

Describe the suitability of using the electoral register rather than a questionnaire.

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1 Using a numerical example, explain the differences between data and information.

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- 4 A bank stores a database of all its customers. The first field in the database is the account number. The account numbers are different for each customer and consist of 10 digits, for example 7382690643.

New accounts are opened in branches of the bank on a regular daily basis.

- (a) Describe **three** errors that a person entering the account numbers of new customers might make. For each one describe an appropriate validation check that would trap the error. A range check or limit check would **not** be suitable. Each validation check must be different.

Error 1

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Validation check 1

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Error 2

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Validation check 2

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Error 3

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Validation check 3

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[6]

When customers open new accounts, the details are gathered together in a transaction file. At the end of the day a branch transmits the transaction file to head office, using the internet.

(b) The transmission of the transaction file is verified using a hash total.

Describe this process, explaining why a hash total is used rather than a control total.

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4 An airline company uses online processing to allow passengers to book their seats.

(a) Describe how online processing is used in this scenario.

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(b) Describe **two** benefits for passengers of using online processing when booking seats.

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Benefit 2

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1 (a) Describe, giving an example, what is meant by the term data.

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Example.....

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(b) Give an example of information and explain why your chosen example is considered to be information.

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2 Identify **four** factors which can affect the quality of information.

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10 Explain how each of these methods keeps data safe from unauthorised users.

(a) encryption

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(b) a firewall on a network

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5 There are various methods of verification.

(a) Describe why checksum verification is used.

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(b) Describe a method of calculating a checksum.

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5 A college uses a database to store student records. Each student has a unique ten-digit student number.

The database stores the examination scores of each student. The maximum mark in each examination is 100.

The database already has presence, existence and integrity validation checks built in.

Describe **three** additional validation rules, using student number and examination scores as examples. For each rule, give a different reason why it can still allow inaccurate data to be entered.

Rule 1
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Reason 1
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Rule 2
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Reason 2
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Rule 3
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Reason 3
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[6]

5 A scientist is going to use direct and indirect data sources to study the effects pollution is having on farmland.

(a) Explain what is meant by direct data sources. Your answer must include examples linked to the scenario.

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(b) Explain what is meant by indirect data sources. Your answer must include examples linked to the scenario.

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6 Explain the need for both validation and verification to ensure data is accurate.

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2 The quality of information can affect the usefulness of decisions made by businesses.

Describe how the following factors can affect the quality of information and the quality of the decisions made.

(a) age of information

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(b) relevance of information

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8 A furniture company uses databases in its warehouse to help with stock control.

(a) The company has a Product database which contains the fields shown in the table.

Field Name	Data Type	Example of Data
Number_in_Stock		199 (Number_in_Stock can vary between a minimum of 0 and a maximum of 250.)
Location		07.44.03
Reorder_Level		250
Price		\$203.50
Supplier_Telephone_No.		07700900819

Complete the Data Type column in the table by writing the most appropriate data type for each field. For any numeric fields, specify the numeric data type. [5]

- (b)** State and describe appropriate validation checks that could be carried out on data as it is entered into the Product database. You can describe different validation checks for the same field.

[6]

- (c) As well as the Product database, the company has a Customer database. Describe how the two databases can be combined into one relational database.

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[8]

4 Validation and verification can be used to check the accuracy of data.

(a) Describe the differences between validation and verification. Your answer must explain the purpose of each.

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(b) Describe each of the following:

(i) range check

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(ii) double data entry

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5 Describe the use of a public key and a private key in asymmetric encryption.

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Explain, with examples, how these items of data can become information.

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Mark scheme

- Visual checking is used to verify that the data is the same as the source document.

Describe, by using an example, how visual checking could still result in errors in the leaflet.

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- The system already uses presence checks on the student number and examination scores.

- student number
- examination scores.

[8]

8 Describe the use of Internet Protocol Security (IPsec).

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1 A teacher is collecting information from her students about what items they recycle.

Describe **two** factors that can affect the quality of the information gathered in this scenario.

Factor 1

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Factor 2

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1 A company is going to carry out an investigation into shopping habits. The company will get data from indirect data sources.

(a) Describe the benefits of getting data from indirect data sources.

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(b) Describe the drawbacks of getting data from indirect data sources.

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11 A college database contains details about students. Part of the database is shown.

StudentID	FirstName	FamilyName	PhoneNumber
AB37689	Mohammed	Abbas	016324908367
AH83965	Jonathan	Olsen	016328332901
BV72189	Suzanne	Usher	016325599100
DY64817	Hua	Owyang	016323652331
KT52964	Isabella	Mancini	016328899477
PS54200	Ben	Quansah	016329677901

Identify and describe **two** validation checks that are suitable to use on the StudentID field.

Validation check 1

Description 1

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Validation check 2

Description 2

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[4]

[8]

9626/13 May/June 2025 · Question 1

[Mark scheme](#)

1 Observation is one method of collecting direct data.

Identify and describe **two** other ways of collecting direct data.

Way 1

Description

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Way 2

Description

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[4]

9626/13 May/June 2025 · Question 2

[Mark scheme](#)

2 Describe the use of TLS/SSL.

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9626/11 and 9626/13 Oct/Nov 2025 · Question 1[Mark scheme](#)

- 1** (a) Describe what is meant by data.

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- (b) Describe what is meant by information.

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9626/11 and 9626/13 Oct/Nov 2025 · Question 2[Mark scheme](#)

- 2** A hospital nurse records direct data about her patients every day.

- (a) Describe the benefits of using direct data in this context.

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- (b) Describe the drawbacks of using direct data in this context.

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2 Describe how a control total is used to verify data.

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7 A company wishes to gather some information from employees.

(a) Describe the process of organising face-to-face interviews.

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(b) Justify the use of a relational database to store the information gathered.

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[6]

Mark scheme

- Explain why this data is **not** information.

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- 10** A list of some field names from a database is shown. For each field name, identify the most appropriate data type.

(a)

Field name	Most appropriate data type
employee_name	
hourly_rate_of_pay	
was_employee_born_in_france?	

[3]

- (b) A presence check has already been run on each field of the database.

For each of the following fields, identify and describe **one** other type of validation that can be used. You should only use each type of validation once.

- (i) hourly_rate_of_pay

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- (ii) date_of_birth

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- (c) Justify why a field holding the age of each employee has been set as a calculated field.

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1 Data may be described as direct or indirect.

(a) Describe **two** differences between direct and indirect data.

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(b) Identify and explain **one** circumstance in which it would be better to use indirect data for a project.

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- 1 A businesswoman is opening a shop which will sell sports goods. She has decided to research into the spending habits of potential customers.

(a) (i) Describe **one** source of direct data she could use for this purpose.

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(ii) Identify **one** source of indirect data that she could use for this purpose.

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(b) With reference to **both** relevance of data and age of data, explain why the information obtained from the direct data source may produce better quality information.

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The businesswoman has decided to use indirect data for her research into the spending habits of potential customers.

- (c) Identify and explain **two** possible reasons, other than relevance of data and age of the data, for her decision to use indirect data for this purpose.

reason 1

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reason 2

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[4]

4 In a school, a student needs to include sections from books in an essay he is writing.

Each section must be an exact copy of the section in the source material.

- (a)** Compare the use of visual checking and double data entry to verify the sections used. You must include similarities and differences in your answer.

[4]

In a workplace, a file containing employee details, including employee ID numbers, is going to be transferred by File Transfer Protocol (FTP) to a manager's computer. The integrity of the file has to be checked after it has been downloaded.

- (b)** Identify and describe **one** method of verification, other than visual checking and double data entry, that can be used to check the integrity of the file downloaded to the manager's computer.

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- 5 A school has paper-based records of all students that have attended the school since 1993.

These records include:

- home addresses and contact details
- the grade each student achieved in each of their end of year exams
- the years when they joined and left the school.

The school is going to create and use a single database to store these records.

- (a) Compare the use of flat file database structures and relational database structures to store these records.

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The database will include validation rules.

- (b) Identify and describe **one** validation rule that could be applied to the database. State why it is important.

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Mark scheme

- 1** Describe, with an example, the differences between data and information.

[3]

9626/13 May/June 2026 · Question 3

[Mark scheme](#)

3 Personal data is held on a hard disk.

Explain **two** benefits of encrypting this hard disk.

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9626/13 May/June 2026 · Question 4

[Mark scheme](#)

4 A shop owner needs to collect data about the shopping habits of potential customers. This data needs to be high quality.

(a) Identify and explain **one** reason why this data must be of high quality.

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The shop owner has decided to use an in-depth report that he has found in an online newspaper. This report contains data about national spending habits.

- (b)** Evaluate the benefits and drawbacks of using this data to identify shopping habits of potential customers.

[8]

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 1

[Back to question](#)

Question	Answer	Marks
1(a)	Four from: Data on its own has no meaning and there is no meaning to this data This data is just raw facts and figures Data has to be processed to become information and there is little evidence that any of this data has been processed as it has no meaning Data needs to have a context in order to become information/we don't know the context for this data... ...description using an example of why there is no meaning giving two alternative contexts Data can be in the form of numbers, characters, symbols as shown in the example	4
1(b)	Three from: 1989 could be a year, such as the year somebody joined a company Paolo could represent somebody's first name √ could be the square root symbol used in Maths ₹ could represent a currency symbol such as is used in India ≠ could be the not equal to sign used in Maths OR could be a Boolean operator used in IT/Computer Studies	3

Mark scheme · 9626/12 Feb/March 2022 · Question 6

[Back to question](#)

Question	Answer	Marks
6(a)	Three matched pairs (all must be different): A length check could be performed on the Student_number data Each data item must be exactly 5 characters long/no more and no less than 5 characters A format check could be performed on the Date_of_birth data Each date must consist of two digits followed by a slash, followed by two digits, followed by a slash, followed by four digits A length check could be performed on the Date_of_birth data Must be exactly 10 characters A consistency check could be performed on the Student_number and Date_of_birth data The first two digits/year group must relate to the Date_of_birth	6

Question	Answer	Marks
6(b)	Six from: Four from: Common errors when copying data are transcription and transposition errors Verification would pick up a transposition error, but most validation checks (except a check digit) would not Verification would pick up transcription errors, but validation might not Verification is checking that data that has been/is being entered into a computer has been <u>copied correctly</u> from the data source Validation is checking that the data entered is reasonable/sensible <u>Neither</u> method checks that data is accurate/correct Examples from: A format check might show that the Date_of_birth is in the correct format of dd/mm/yyyy but verification would pick up errors... ...Jonathon Odoki's Date_of_birth, for example, might have been incorrectly copied as 12/07/2004/month and day have been swapped A length check might show that the Student_number is the correct length but verification would pick up a transposition error... ...For example Sven Johansson's Student_number could have been copied as 09312/two digits have been swapped A length check might show that the Student_number is the correct length but verification would pick up a transcription error... ...For example Sven Johansson's Student_number has been copied as 09S32/text has been entered instead of a digit A consistency check might show that the Date_of_birth is consistent with the Student_number but Manjit Sangherra's Student_number may have been copied as 11102... ...and Date_of_birth as 11/03/2006 Allow follow through from part (a)	6

Mark scheme · 9626/11 May/June 2022 · Question 1

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Question	Answer	Marks
1	Four from: It is collecting the data and then processing it/processing it all in one go/all at once / stored until the system comes online to process the data in one 'batch' The customer purchases/repayments/orders are collected into batches and each batch/orders are processed as a whole Customer orders processed when the computing resources are less busy Batches/orders can be stored up during working hours and then executed/processed during the evening/weekend Transaction file is used to update master file Transaction file of customer purchases/orders is kept for later processing – weekly/monthly Master file of customer details is only updated weekly/monthly Transaction file is used with master file to produce customer orders at the end of the month	4

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Question	Answer	Marks
3	Eight from: Encryption is the scrambling of data... ...it converts plaintext to ciphertext... ...into meaningless groups of symbols... ... can't be understood by unauthorised people/read without the decryption key... ...without encryption messages can be stored as plain text, making it easy for anyone to read and understand Encryption keeps personal data secure... such as credit card numbers and personal information from computer hackers If personal information is intercepted, but it can't be understood, it is of no use... ... the act of hacking in this case is pointless It does not prevent hackers intercepting personal data but it prevents them from understanding it Two types of encryption asymmetric and symmetric... ...symmetric and asymmetric use a private key but asymmetric uses a public key as well /symmetric uses the same key to encrypt and decrypt but asymmetric uses different keys... ...symmetric uses a private key to encrypt and decrypt /symmetric is where both sender and receiver use a private key... ...asymmetric uses a public key to encrypt and a private key to decrypt /sender uses a public key and receiver uses a private key Can only decrypt using a decryption key/only authorised users have the decryption key... ... Symmetric is a faster method than asymmetric... ... Symmetric is less secure than asymmetric Data is encrypted using an encryption key Must be a proper analysis to gain full marks Max. six marks if bullets/list of points Must have expansions to be a proper analysis	8

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Question	Answer	Marks
4	Seven from: The computer searches for the bar code number 5022476130067 in the product file When the matching record is found, the number in stock, 131, is reduced by one to 130 The number in stock, 130, is now compared with the re-order level, 130 If the number in stock is equal to the reorder level... ... which it is, then the computer creates an order automatically The computer reads the reorder quantity, 250, needed for the product It reads the supplierID, B671841, for Oaters It searches the supplier file for the record corresponding to the supplier ID, B671841, read from the Product file The order/e-mail is automatically sent to the supplier... ...using the supplier's contact details from the Supplier file, js@mail.co.ke	7

Question	Answer	Marks
8	Eight from: <i>Advantages</i> It may allow a larger set of data to be examined... ...direct data gathering could have limitations due to the availability of the people being interviewed... ...using banks the sample could be increased giving rise to greater confidence in the findings Information can be of a higher quality as banking data will have already been collated and grouped into meaningful categories Joanne does not have to read through poorly written responses to questionnaires/her interview transcripts... ...to create the data source <i>Disadvantages</i> It may cost a lot of money to pay the banks to obtain the data... ... whereas people who are questioned will not need to be paid The various purposes for which data was collected originally may be different to the purpose of the current study... ... there may be much more data than needed... ...unnecessary data may need to be filtered out There may be <u>sampling</u> bias There may be coding difficultiesdifferent banks being used means there will be differences in the content ...there may be a lack of standardisation across the data required ...different banks will have different amounts of data If it has already been coded the coding may be difficult to understand Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

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Question	Answer	Marks
1	Five from: By interviewing the family, the students will obtain up to date information/census is two years old/census data may be out of date More/detailed information will be obtained as the census may not contain all the information required/has limited information/fewer details They are able to collect only as much information as they require/the census may include information they do not needmight take time to filter extra information out It may be cheaper to visit the streets than the school paying transport costs to get the students to city hall It may be too time consuming to collect direct data / to go to every house in these streets / only need to make one trip to city hall The study might be limited as it is only five streets/the census would allow them to extend the survey Residents might not be at home when the students call/nobody at home would not be a problem with the census Data may take time to collate and present in a meaningful way/census data is already collated and presented so it is easier to find the required information	5

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Question	Answer	Marks
2	Four from: When a file is written to the disk, it can be automatically encrypted by encryption software Full disk encryption is when the data is encrypted as soon as it is saved to the hard disk When a file is read from the disk the decryption/encryption key/password is needed to decrypt it When a file is read from the disk, the software automatically decrypts it... ... while in some types of hard disk encryption leaving all other data on the disk encrypted (All) the files on the disk are encrypted/ It's the encryption of data on that disk... ...except in some cases the operating system Whether the disk stays in the computer or moved to another computer it is still encrypted Only the person/computer with the secret/decryption/encryption key/password can understand the data on the disk/data can't be read without the decryption/encryption key/unauthorised people cannot read the data	4

Question	Answer	Marks
3(a)	Four from: For even parity: Three max from When data is transmitted the sending device/computer counts the number of ones in each byte Each byte is checked individually The parity bit is used to make sure the number of ones is even If the number of ones is even it sets the parity bit to 0 (and adds this on to the end of the byte) If the number of ones is odd it sets the parity bit to 1 (before adding it on) Every byte of data should have an even number of ones The receiving device/computer checks each byte to make sure that it has an even number of ones If the number of ones is odd there has been an error during the transfer of data Two max from In this example data is not being transferred from one device to another/data is not being transmitted She is only entering data to the computer so will need to use an alternative verification method Accept answers which are odd parity answers (substitute odd for even in points above)	4
3(b)	Four from <i>Similarities</i> Both involve comparing data that has been/is being entered with another copy Both involve checking that data <u>has been entered</u> correctly not that it is correct/pick up errors made during data entry <i>Differences</i> Visual verification is carried out by a human/user/yourself Double data entry is carried out by the computer/ computerised Visual verification involves comparing the entered data with the original source Double data entry involves comparing two versions of the entered data Visual verification will enable the user to see mistakes if they have been made With double data entry computer will alert user to the fact that a mistake has been made and they decide which version has been copied correctly/freezes keyboard to prevent data being entered until error is corrected Must have at least one similarity and one difference to gain full marks	4

Question	Answer	Marks
9(a)	<pre> READ first record in transaction file READ first record in old master file WHILE not end of transaction file IF transaction file WorkerID = master file WorkerID THEN WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) WRITE updated master file record to new master file READ next record in transaction file ENDIF READ next record in master file ENDWHILE </pre> Six from: Correct calculation – RateOfPay*HoursWorked – 1 mark Correct calculation - WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) – 1 mark Correct update – 1 mark Read next transaction file record... – 1 mark ...immediately before ENDIF – 1 mark ENDIF– 1 mark Read next master file record... – 1 mark ... immediately after ENDIF – 1 mark ENDWHILE at end of algorithm – 1 mark	6

Question	Answer	Marks
9(b)	Eight from: <i>Advantages</i> Batch processing does not require constant use of the computer when it might be needed for other tasks... ...the payroll only needs to be run at the end of the week/month... ...whereas real time processing would require computer to be available all the time... ...batch processing/payroll can be run at convenient times for the company/when the computing resources are less busy/when there is little demand for computer resources/can be run overnight Real-time processing requires immediate processing of data which is unnecessary with payroll... ...as hours worked would only be known at the end of each day so no point in using real-time processing Real-time processing is not usually capable of processing large volumes of data such as running a payroll whereas batch processing is Batch processing systems only need lower specification computers... ...whereas real-time processing needs more complex computer systems Online processing requires more hardware making it more expensive than batch processing Batch processing only needs a few employees to enter the data... ...so total wage bill is less... ...whereas online processing systems can need employees to be constantly using the system In batch processing systems there are fewer/no transcription/update/data errors... ...than human operators would produce in an online system <i>Disadvantages</i> With batch processing there is a time delay between collecting the input data and getting an output/data is not processed until the time specified With batch processing the information is only up to date after the master file has been updated by the transaction file... ...whereas in real-time processing information is always up to date With batch processing systems errors are only revealed during processing/after batch process is complete which can be overnight and so cause delays/errors cannot be corrected until the batch process is complete... ...in online processing errors are revealed, and can be acted upon, immediately Users cannot terminate a process during execution/have to wait until execution is completed Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

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Question	Answer	Marks
1(a)	Data that is collected for a specific purpose or task/ for that purpose only/gives us data that is often called 'original source data'	1
1(b)	Four from: Questionnaires could be distributed amongst neighbours/local people asking about the level of noise Interviews could be carried out with neighbours/local people (asking about the level of noise) He could observe traffic, keeping a count of traffic coming down each street He could use sensors to detect the flow of traffic going down each street He could use sensors to measure the level of noise in each street	4

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Question	Answer	Marks
6	Six from: Transport Layer Security (TLS) is used for applications that require data to be securely exchanged over a client-server network... ...such as web browsing sessions/ file transfers The server shows/sends its SSL/TLS/digital certificate to the client /the client requests server's SSL/TLS/digital certificate To open a SSL/TLS connection, a client needs to obtain the public key The public key is found in the server's digital certificate The SSL/TLS/digital certificate authenticates the server to the client/identifies the server The client then carries out a number of checks to make sure that the certificate was issued by a trusted CA... ...is in date and that the server is the legitimate owner of the public and private keys For the client to access the server the client and server must carry out a SSL/TLS handshake... ...the client tells the server what version of SSL/TLS it uses... ...and a list of the different types of encryption that it is able to use The client tells the server that it wants to set up a communications channel Handshaking occurs before the transfer of data can take place The server tells the client the type of encryption it has chosen from the client's list	6

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Question	Answer	Marks
3	Up to six marks available for three matched pairs e.g. • Delete/remove a (master file) record (1) • To remove duplicates//Where a record is no longer needed (accept examples) (1) • Amend/change a (master file) record (1) • E.g. A worker moves department/gets a pay rise/changes name (1) • Add/append a (master file) record (1) • A new worker joins the company (1)	6

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Question	Answer	Marks
4	Six from: • Chip is read/checked ... (1) ... to make sure it has not been reported stolen, is in date and is a valid card number (MUST be two of three) ... (1) ... If not message appears saying the card is invalid (and the transaction does not continue) (MAY BE AWARDED WHERE ONLY ONE OF PREVIOUS THREE EXAMPLES HAS BEEN GIVEN) (1) • PIN entered by the customer is compared with that stored on the chip (1) • If PIN is verified the transaction continues ... (1) ... otherwise (error) message will appear ... (1) ... customer is asked to re-enter PIN (1) • If after three attempts the PINs do not match the transaction is rejected (1) • Supermarket computer contacts customer bank (to find customer's records) to see if there are sufficient funds in the customer's account (1) • If there are sufficient funds in the customer's account transaction is completed (1) • If there are insufficient funds in the customer's account transaction is rejected (1) • Amount of the transaction is deducted from customer's bank account (1) • Amount of the transaction is credited to supermarket's bank account (1)	6

Mark scheme · 9626/11 Oct/Nov 2022 · Question 5

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Question	Answer	Marks
5	Five from: • Often called public-key encryption (1) • Uses two different keys// one <u>public</u> and one <u>private/secret</u> (key) (1) • No need to transfer a key (with the message) (1) • The public key/key held by the person doing the encryption is used to encrypt the data (1) • The corresponding private/secret key is used to decrypt the data (1) • The public key is published to everyone (1) • The private key is kept secret (1) • Anyone with a copy of the public key can encrypt information (1) • Only the private key holder can read the information (1) • It is not possible to deduce the private key from the public key (1)	5

Mark scheme · 9626/11 Oct/Nov 2022 · Question 11

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Question	Answer	Marks
11	Max TWO from: e.g. • Data has been collected for a particular/different purpose ... (1) ... but then that data is used for something else (1) Max TWO from: e.g. • The Electoral Register is a list of adults who are entitled to vote in a local or national election (1) • Customers in stores often provide personal information when they buy products (1)	4

Mark scheme · 9626/12 Oct/Nov 2022 · Question 1

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Question	Answer	Marks
1	MAX four marks: Max three from: - Information is data which has a context/meaning/has been processed// this data can become information if meaning/context is added (accept any indication that meaning or context is key to data being information) (1) This data/it has no context/meaning//it doesn't explain what each symbol means (must clearly refer to the set of data here)//it is raw (1) - Any valid context//The context for this set of data could be that it can be used in search criteria in spreadsheets ... (1) ... the symbols would then be given meaning (1) Max two from: 1 mark for 2 correct meanings (1) OR 2 marks for 4 correct meanings (2) > means greater than < means less than = means equal to >= means greater than <u>or</u> equal to <= means less than <u>or</u> equal to	4

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Question	Answer	Marks
5(a)	Three matched pairs: - A <u>length check</u> could be performed (on the data) in the ISBN field (1) Each data item must be <u>13</u> characters/numbers/digits long (1) - A <u>lookup check</u> can be performed (on the data) in the Book type field (1) Each data item is compared with the lookup list of P and H to see it exists//The code must be P or H only (1) - A <u>range check</u> could be performed (on the data) in the Copies field (1) Each value must not be less than the minimum/2 and must not be greater than the maximum/5//between <u>1 and 6</u> (1)	6

Question	Answer	Marks
5(b)	Three marks available: • Generic statement about why both required (no reference to fields). MUST be more than a statement about what each does. (1) • Verification would identify if P had been typed in the Book_type field for the first record//if it was relevant (accept any letter and book combination from the sample) (1) • A lookup check/validation would accept P or H (1) • Verification would identify if (e.g) 5/wrong number had been entered (into the first record) (1) • A range check/validation would accept 5 (1) • Verification would identify if two digits had been transposed in the ISBN field//the ISBN number was incorrect (1) • Length check/Validation would accept any 13 characters (1)	3

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Question	Answer	Marks
8(a)	Four from: • When a message/information is intercepted//accessed it is unreadable/ can't be understood ... (1) ... so, therefore, it is useless (1) ... to a hacker (accept alternative examples of 3rd party or just 'third party') (1) • It protects customers when they bank/shop online/any other suitable example from any area of use (MUST BE protecting data) ... (1) ... as the data cannot be used for identity theft (1) • Data can only be decrypted by the receiving computer if it has the private/decryption key (1) Max. three marks if bullets/list of points	4

Question	Answer	Marks
8(b)	Up to FOUR marks available: Advantages (MAX 3) • It is a faster process//quicker to encrypt compared to asymmetric (1) as it has less mathematical complexity (1) • It requires less computational power than asymmetric (1) Disadvantages (MAX 3) • Does not have a different key for encryption and decryption//sender and receiver use the same key (1) therefore only one key needs to be stolen (1) ... making it less secure (1) • The encryption key cannot be generally published (1) • The encryption key must be sent to the receiver of the message (1) • Key could be intercepted/stolen (by a hacker) whilst being sent ... (1) ... who can then decrypt any message encrypted by the sender (accept any reasonable implication) (1) Must have at least one of each to gain full marks Max. three marks if bullets/list of points	4

Question	Answer	Marks
1	Six marks available: - Identifies Accuracy as a factor (MUST be more than just the word 'accuracy') (1) - Identifies Relevance as a factor (MUST be more than just the word 'relevance') (1) <i>Accuracy</i> - Information must be accurate to be considered of good quality (1) - If the collected data is inaccurate then the information produced will also be inaccurate (1) - The method of collection can affect the accuracy of the collected data and thence information (1) - If questions in an interview/on a questionnaire are not phrased correctly this leads to inaccurate data being collected (then the information produced will also be inaccurate) (1) - If data is then entered incorrectly it will lead to inaccurate information being produced (1) - Example of inaccurate information/data (1) <i>Relevance</i> - The relevance of the data relates to the situation or problem being studied (1) - Relevant information is information which meets the needs of the user/relevant to its purpose (1) - It should not concentrate on one aspect of the problem (1) - It should not relate to a different area to the one being studied (1) - Should not have too much information ... (1) ... user will spend a lot of time searching to find relevant information (1) - In order for information to be useful it needs to be complete ... (1) ... if only some of the information required is present then it is very difficult to solve a problem/complete a task (1) - Example of irrelevant information (1) MUST be at least one for each side for full marks	6

Question	Answer	Marks
2	Five marks available: • A real-time system has to respond to/process inputs//inputted data instantaneously (1) • An air conditioning system has to deal with inputs continuously ... (1) ... unless the user switches the system off (1) • This type of processing is usually found in systems that use sensors (1) • The user interface of a real-time system may use specialist input devices to provide data input ... (1) ... such as a keypad/touchscreen (1) • Usually deals with small amounts of data//does not deal with a lot of data (1) • Real time systems are such that the output affects the input (1)	5

Question	Answer	Marks
1(a)	ONE from: - Information should be/that is free from errors/mistakes (1) - Information should be/that is true/correct (1) - Information that is provable/proven to be correct (1)	1
1(b)	SIX from: - People might not answer the questionnaire truthfully (1) - People might answer the interview truthfully (1) - People may be biased (1) - The questions might be badly phrased (1) - Interviews can clarify the answer (1) - Interviews can ask questions based on previous answers (1) - Questionnaires <u>cannot expand</u> on what the question means (1) - Interviews include open ended questions that are hard to quantify (1) - In a multi choice type question there may not be a sufficient number of alternatives//answers restricted to those provided (1) - There is the possibility that the people collecting the data may make errors when collecting it (1)	6
1(c)	ONE from: - User is in communication with a central computer (1) - Where a process is carried out on the internet/WWW (1)	1
1(d)	Five from: - Go to the website (1) - Log on (by entering username and password) (1) - Choose goods category (1) - Choose/select item (to examine the product) (1) - Add (items) to shopping basket/cart //click on buy button//pick up item (1) - Check good in basket/cart//make changes to goods in basket/cart (1) - Click on/proceed/go to checkout (1) - Confirm billing address (1) - Confirm/enter delivery address (1) - Choose method of delivery (1) - Select/use payment method (1) - Type in credit card/debit card number (1) - Enter card security code/last three digits of CVV/CVC/CSC (1) - Confirm order//click checkout button//click pay (1) (NO further marks to be given after this mark given, unless candidate clearly states that the action happens before this point)	5

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Question	Answer	Marks
5	FOUR from: • (Using data encryption is) a way to keep their data secure//to protect their information from a hacker//to safely exchange data (1) • Information needs to be impossible to understand/unreadable/indecipherable (1) • Information needs to be useless to a 3rd party if intercepted/hacked/they should not have it (1) • Only authorised people/people with the key can understand the information (1) • Encrypted data needs to be decrypted which is hard/impossible (without the key) (1) • To make hacking not worthwhile (1)	4

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Question	Answer	Marks
1(a)	One from: • Data is scrambled/jumbled up (1) • Converting information so that it is impossible to interpret/understand (1)	1
1(b)	Up to THREE marks: e.g. • Can be used for identity theft (1) plus one example (1) • Can be used for identity fraud (1) plus one example (1) • Company secrets could be sold by hackers to rival companies (1) • National secrets could be used by rival powers (1) • Secrets/personal data could be used for blackmail purposes (1) • Private details could be used to pass (bank) security tests (1st) – Accept any suitable extension for passing security tests (1) e.g. – And so access (e.g.) bank accounts (1)	3

Question	Answer	Marks
5(a)	Validation and sorting (1)	1
5(b)	Eight from: • First record//2210 in the transaction file (1) • First record in the master file/1965 is read/checked (1) • These (previously identified) two records are compared (ALLOW FT if attempt at identifying both records) (1) (Award 'the' instead of 'these' where MP1 AND MP2 given) • Records don't match (1st) – computer writes master file record to new master file (1) • Next record of master file/2210 in the master file is read/checked (1) • Transaction file record/2210 is compared with this record (1) • It matches so the Numbersold is subtracted from NumberinStock (1) • Using NumberinStock of 12000 from master file (1) • Using Numbersold of 6500 from transaction file (1) • NumberinStock is now 5500 (1) • Processed record/NumberinStock is written to new master file (1) • Next record/3356 in the transaction file is read/checked (1) • Next record/2976 in the master file is read/checked (1)	8

Question	Answer	Marks
11	Eight from: - Any correct definition of direct data sources, such as: - Direct sources are where Isabella asks people in person//primary source of data (must be an attempt to define) (1) - Can gain/gather qualitative AND quantitative data (1) - Data gathered is more up to date/recent (1) - Data is (guaranteed to be) relevant to the study/question (1) - Interviewer can base further questions on answer given (1) - Interviewer can ask a more in-depth/follow up questions (1) - Can interpret body language (1st) - such as (e.g.) facial expressions//nervous movement of hands (1) - Interviews tend to be taken seriously by people (1st) - whereas questionnaires may be ignored (1) - It ensures the data is relevant to the study//you get a better view of what is going on//get a better understanding (1st) - whereas indirect sources are not specific to the study (1) - You are in control of the sample size//can be as big/small as needed (1st) - whereas Isabella has no control over the amount of data collected from an indirect source (1) - The source of the data is known exactly (1st) - making it easier to judge its reliability (1) - Isabella could sell/pass/share the data on (to interested companies) (1st) - indirect data is not hers to sell/pass/share (1) MAX 6 marks for a bulleted list	8

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Question	Answer	Marks
1(a)	Five from: • It would now be used for a different purpose to that for which it was originally collected (1) • Electoral register only shows adults (accept does not show children) (1) • Only shows people who are registered to vote (1) • The data on the electoral register might be out of date//not have been updated (1) people on the electoral register might no longer live in the area (1) ... people new to the area might not be included (1) ... new homes may have been built (1) ... people may have died (1) • Not all people who live at the same address are from the same family (1)	5
1(b)	Award ONE mark for ANY indirect data source: e.g. • Businesses collecting personal information when used by third parties (1) • Census (data) (1) • A newspaper (1) • An encyclopedia (1) • Medical records (1) • Museum archive (1)	1

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Question	Answer	Marks
2	Five from: • One mark for a description of encryption (1) • Often called secret key encryption (1) • Uses a single/the same key (to decrypt/encrypt the data) (1) • Any awareness that it is a private key (1) • This (private/secret) key must be shared between the sender and the receiver (1) • The encryption key is sent to the recipient (1)	5

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Question	Answer	Marks
3	Five from: MAX three if only batch processing OR online processing discussed. • Data is grouped into batches (in batch processing) (1) • With online processing, data is processed as quickly as possible (1) • With batch processing there is a delay between the collection of data and the actual processing (1) • With batch processing, the data is entered and processed altogether in one batch by the computer (1) • With online processing the user tends to be in direct communication with a central computer (1) • Online systems require more hardware/input devices/workstations than batch processing systems (1) • Batch processing uses very little processing time compared to online processing (1) • In online processing errors are revealed immediately (1) • (whereas) Errors in batch processing errors are revealed much later/only when processing takes place/overnight/when batch process is complete (1) • Batch processing can be done when the system is not busy//scheduled to a specific time (1) • (whereas) online processing has to be active for the whole day (1) • For batch processing, data is only accurate at the time of processing (1) • (whereas) online processing, the data is accurate all the time.	5

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Question	Answer	Marks
1	Four from: • Automatic collection/storing of data (1) • Using a computer and sensors to collect data (1) • This data is analysed/processed (1) • The results are output in the form of graphs/charts/tables (1) • Data is collected over a period of time ... (1) ... continuously/at regular intervals (1) • The analysis/processing usually requires special software (1) • Two of the following examples of physical variables logged: temperature sound light pressure (1)	4

Question	Answer	Marks
2	Five from: • ONE of: SSL is Secure Socket Layer OR TLS is Transfer Layer Security (1) • Enable encryption in order to protect data (1) • To protect data whilst being transferred (accept close alternatives to 'transferred') (1) • To encrypt and decrypt data sent/received by web users (1) • Provides secure connection between web browser and website/client and server (1) • Authenticate the people/companies/users exchanging data (1) • Ensure the integrity of the data to make sure it has not been corrupted/altered (1) • Enable credit/bank card payments to be made securely ... (1) ... by setting standards for the storage/transmission/processing of bank card data (1) • Improve customer trust/more likely to do business with a company using SSL/TLS protocol on its website (1) • Enable users to securely complete online transactions (1) • Enable the creation of a secure connection between a web server and a browser (1) • Prevent eavesdroppers from trying to intercept data (1) • To provide security of communication in a network/secure transmission/sharing of data (1) • Encryption protocols enable credit card payments to be made securely (1)	5

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Question	Answer	Marks
3(a)	Three from: - Data is grouped (1) - A group of data is processed all at once//in one go (1) - Jobs/batches are set up so they can be run to completion without human interaction (1) - Data is processed at a set time (1) Takes place when the computing resources are less busy (1) - With batch processing there is a delay between the collection of data and the actual processing (1)	3
3(b)	Three from: - The response time of the computer must be immediate (1) - Real-time responses are in the order of milliseconds, and sometimes microseconds (must be more than 'is fast', but this can be expanded upon)//real time processing is immediate (1) - Real time means that the inputs are processed and produce an output which in turn affects the input (1) - The processing involved in a rocket guidance system has to be continuous, so the process is never ending (1) - An example of consequences (1) - An example of benefit (1)	3

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Question	Answer	Marks
2	Three from: - Validation is always carried out by a computer whereas verification can be carried out by a computer or by a human (1) - Verification is checking that data that has been/is being entered into a computer has been <u>copied correctly</u> from the data source (1st) - whereas validation cannot tell if the data has been copied incorrectly or not (1) - just checks that the data input is sensible/reasonable/follows a set of rules (1) - Validation is checking that data is sensible/reasonable/follows a set of rules (1st) - whereas verification cannot tell if the these rules have been followed/data is sensible/reasonable (1) - just checks that data input has been copied correctly (1) - Verification, (unlike validation) cannot check whether data is sensible or not (1) - Validation will pick up that data is invalid when collected but verification can only check that the data has been copied/transferred accurately (1) - Verification can pick up a transposition error but most validation checks (except a check digit) would not (1)	3

Question	Answer	Marks
4	Five from: • HTTP protocol with SSL/TSL encryption (1) • Any information transferred from the client to the server/server to the client is encrypted (1st) • enabling the secure transmission of (confidential) data (1) • As part of the handshake (between the web browser on the client computer) a session key is created (1) • This session key is encrypted using the public key and sent to the server (1) • All data sent between the client/browser and server is encrypted using the session key (1) • Session key generated using asymmetric encryption (1) • symmetric encryption is used to encrypt all further communications (1) • Each separate session requires a new session key to be created (1)	5

Question	Answer	Marks
11	Eight marks available: Advantages max 6 • Orders are/can be processed straight away (1st) • so process of filling/completing orders can start sooner (than for batch processing) (1) • With online processing errors are revealed immediately (1) • With batch processing systems if there is an error it is only revealed when processing takes place (1st) • Delay in finding the error (1) • If the database needs to be interrogated details can be retrieved immediately (1st) • Any example of why immediate interrogation would be required – e.g a sales invoice/order may be required when a customer queries a bill (1) • With online systems they are available whenever necessary (accept converse argument) (1) • Data is accurate all the time (1) Disadvantages max 6 • Extra hardware requirements/input devices/workstations of an online processing system can make it more expensive (than batch processing) (must clarify at least one process for the comparison) (1) • Online processing has to be active for the whole day (1) • There must be workers available (to enter data) at all times (1st) • more employees are needed (1) • company wage bills can be higher (1) • Batch processing systems only require a limited number of employees to enter all the data (1) • It is more expensive than batch processing as it uses much more computer processing time to update files (1) • With batch processing, processing occurs at a time convenient to the employees responsible for entering the data (1) • Batch processing only uses computers when the demand on the system is low (1st) • therefore entering data does not slow the network down (1) Max. six marks if bullets/list of points	8

Question	Answer	Marks
1	Max five from: • Verification (only) checks the data is copied correctly (1) • Validation checks/ensures that data conforms to a set of rules//is sensible/reasonable (1) • (Although copied correctly) the original data may have errors in it which could be picked up using validation (checks) (1) • (Although fitting the rules) the copied data may have been copied incorrectly, which would have been picked up by using verification (checks) (1) • Both checks are needed to cover mistakes that the other would not identify (1) Max four from: • (In record 1) 13/06/2019 has been copied incorrectly (as 13/06/2020) this would have been found using verification//example of verification (1) • Validation checks would not have picked up the error in record 1/identified error as it is in the correct format and length (1) • (In record 3) Validation/length check would have checked that the ID number was of the correct length//identified that the ID number had too many digits/was too long (1) • (In record 4) Validation/a range check on the Rate_of_pay field would have accepted only values between 15 and 50 (1) • (In record 4) Validation/Limit check accepting only values equal to and below 50 would have identified the error in record 4 (1) • In record(s) 3/4 the data has been copied correctly so verification would not have picked up this error (1) • Format check can be used on date joined field to check that it is dd/mm/yyyy (1)	6

Question	Answer	Marks
5	Five from: • Questionnaire needs to be written/produced/tested (1) • The Electoral Register is a single document//already gathered//immediately available (1) • Electoral Register data is already collated/processed(1) • Data from questionnaires may take <u>time</u> to collate/process (1) • May take too long to distribute questionnaires (1) • She may not have the time/resources available to send questionnaire to everybody/so many people (Award concept of saving time/resources if answer states how time/resources is saved) (1) • She will only need to make one trip to the council office (1) • It may be cheaper to make one trip to the council office rather than pay transport costs to the various parts of town (1) • Electoral Register only lists family members older than voting age//who are allowed to vote/does not include children (1) • Electoral register only holds limited information: name, address, age, gender (two of) (1) • Questionnaires would gather information about the whole family (including younger children) (1) • Questionnaires can ask specific/qualitative/open ended questions (1) • If using questionnaires, need to hire (e.g.) data entry clerks (which would increase costs) (1)	5

Question	Answer	Marks
11	Eight from: One mark for a description of encryption (e.g. data scrambled/converted to cyphertext) (1) Benefits MAX 6 • When data is sent across the internet in an encrypted form it cannot be understood/has no meaning/makes no sense (1st) • for third parties// unauthorised people//people who don't have of a decryption key (1) • and so is of no use (1) • Keeps (personal) data (more) secure/protects data//makes web browsing more secure (1) • Even if data is transferred to many devices/platforms it is still protected (1) • Enables IT departments/businesses to comply with data protection regulations (1) • Reduces the benefits of/motivation to hack(ing)/commit(ing a) crime (1) • Identification of TWO items of data that could be protected (DNA "banking information" – this is TV. Other examples from banking may be correct. (1) Drawbacks MAX 6 • It requires additional processing power to process encrypted data (1) • It takes longer to transmit encrypted data (1) • It takes longer to open/decrypt encrypted data (1) • The client/browser and server must send messages to each other several times before any data is transmitted (1st) • increasing the time it takes to load a web page (1) • It uses extra storage space for the client/the server (increasing costs) (1) • If the private key is lost, it is extremely difficult to recover the data//data may be lost permanently (1st) • data can be recovered by the reissuing of the digital certificate but takes time (1) • Makes it difficult for authorities to monitor criminal activities (1) Max 6 marks for just bullet points	8

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Question	Answer	Marks
1	Three from: - Data consists of raw figures/characters/numbers (which require processing for interpretation) (1) - Data does not have any meaning/context (1) - Information is data that has been given meaning/context (1) - Information is data that has been processed (1) One mark for any example. Example must cover BOTH data and information (1)	4

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Question	Answer	Marks
4(a)	Three matched pairs from: - Might enter an account number with 9 or 11 digits/less than/more than 10 (1st) Length check to make sure that the account number is exactly 10 characters (1) - Might enter a wrong digit (1st) - Use a check digit which creates a new digit by performing a calculation on individual digits of the account number and appending the resulting digit (1) - Might omit one of the account numbers (1st) Presence check would make sure account numbers are entered. (1)	6
4(b)	Three from: - A key number would be calculated (1) - All the account numbers/numbers in the account number would be added together (1st) - To create a single number (1) - The data/file would be transmitted along with the hash total/key number (1) - The hash total is recalculated once received (1) - The original hash total is compared with the new one (1) - If they do not match an error has been made (1) One from: - A control total checks fields that are used for calculations whereas a hash total is used for fields that hold information (1) - The field holds data, and will not be used for calculations (1)	4

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Question	Answer	Marks
4(a)	Four from: - Computers deal with each input and give (almost) immediate/real time responses//Booking is processed in real time/(almost) immediately - Online processing deals with one transaction at a time Once the transaction has taken place the database will be updated immediately/same time as the booking Once booked seat is shown as booked Once the transaction has taken place the computer sends a message/ticket to the user to inform him/her the seat has been booked	4
4(b)	Two from: - Flight won't be overbooked - Duplicate tickets will not be issued/ Seat cannot be booked by anyone else - Guaranteed that will be on the flight - Customers know immediately if they have successfully booked a seat//get immediate confirmation//get immediate advice if there is an issue - Seat is held as yours for a period of time//have time to think about whether to go ahead	2

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Question	Answer	Marks
8	Six available: - These methods/verification and validation on their own/when not used together do not ensure that the data entered is (fully) accurate/correct - Example of both validation and verification in use to achieve accuracy, with clear explanation of why both are needed. Max five from: - Verification is checking that data that has been/is being entered into a computer has been copied correctly (from the data source)//correctly entered - Validation is checking that the data entered is reasonable/sensible/follows a set of rules - Although verification helps to stop users from making mistakes when entering data, it cannot check that the data was originally correct - Validation is needed because although the data might be copied correctly, the original data might be invalid - Therefore, both are needed to make sure that the data is both correctly copied and valid - Valid example of validation technique that could be applied (must be identified as an example of validation) - Valid example of verification technique that could be applied (must be identified as an example of verification)	6

Question	Answer	Marks
9	Seven from: One mark for a description of encryption (e.g. data scrambled/converted to cyphertext) Benefits Max six from: • Encryption makes data unable to be understood/accessed (1st) – for third parties// unauthorised people//people who don't have a decryption key (1) – and so is of no use (1) • Reduces the motivation for interception (1st) – Because data is protected (1) • Allows confidential/sensitive information to be transferred/shared/included in emails (accept converse argument about absence of encryption) • Enables IT departments/businesses to comply with data protection regulations Drawbacks Max six from: • Hackers can still intercept emails (1st) ...data can still be lost (1) • Encryption/decryption/private key must be transferred between sender and receiver • Anyone with the key can understand the email (1st) ...so gives the intended user a false sense of security (1) • If the same key is used repeatedly there is a security issue • Managing digital certificates can become complex and time consuming • Requires more computer processing power • It takes longer to transmit encrypted data • It takes longer to open/decrypt encrypted data	7

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Question	Answer	Marks
1(a)	One from: - It has no meaning (1) - Raw/unprocessed values (1) One from: - any correct example (1)	2
1(b)	One from: - Any example. Must clearly include context or meaning (1). May simply be a description of a type of information. One from: - Processed data (1) - Data + meaning (+ structure) (1)	2

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Question	Answer	Marks
2	Four from: - Accuracy (1) - Relevance (1) - Age (1) - Level of detail (1) - Completeness (1)	4

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Question	Answer	Marks
3	Seven from: Sale • Computer reads the barcode number (1) • Computer searches for this barcode number in the product file/database (1) • If this barcode number is found, then number in stock is reduced (1) Order • Computer compares the number in stock to the re-order level/set value (1) • If the number in stock is not equal (or lower) to the re-order level/set value another barcode is input//systems returns to the start (1) • If the number in stock is equal/lower than the re-order level//a threshold/set value (1st) – the computer creates an (automatic) order (1) • Item is flagged to be reordered (1) • Computer looks up/finds the re-order quantity (for that product) (1) • Computer looks up/finds the supplier (reference) number (for that product) (1) • Computer searches/checks (the supplier file) for the record of the supplier (1) • Computer orders automatically (1st) – using the supplier's contact details. (1)	7

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Question	Answer	Marks
10(a)	Three from: • Data is scrambled (1st) – so it is not understandable (1) – of no use to a hacker/third party (1) • Can only be unencrypted with a key//need the key to unencrypt//access the data (1) • Use of/access to encryption key/software known only to authorised user (1)	3
10(b)	Three from: • Acts as 'barrier' between internal and external networks/internet (1) • Reads (packet) sender's IP address/reads domain of sender (1st) – checks against database/list of IP addresses (1) – checks against database/list of network ports (1) • Permits access/passage of packet if not forbidden/if on allowed list/rules (1) • Drops packet if on forbidden/not on allowed list (1) • Can create logs for analysis (1) • Can create alerts. (1)	3

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Question	Answer	Marks
4	Eight from: Data answers - Transmitted data/file is encrypted (1) so that a third party cannot understand the data//data/file is useless to a hacker (1) - Only the user's computer and the secure/receiving server can recognise the data/file (1) - SSL/TLS use two keys//private and public keys (1) Certificate/process answers (may be answered in any order) - Websites often use (SSL/TLS) certificates to verify their authenticity (1) - Web server is asked to identify itself (1) - The web server sends a copy of the (SSL/TLS) Certificate (1) Certificate Authorities are used to ensure the certificate is real/authentic of the certificate/provides a list of approved certificates (1) - The browser checks to see if the SSL/TLS Certificate is trusted (1) - If the SSL/TLS Certificate is trusted, then the browser sends a message to the web server (1) - If the SSL/TLS Certificate is not trusted, the browser will refuse to open the website (1) and will display a message to the user (1) (IF TRUSTED) The server then responds to the browser with a digitally signed acknowledgement (1) to start an SSL/TLS encrypted session (1) - Session keys are set up (1) - SSL/TLS requires a handshake to be carried out/the browser and the website carry out a (SSL/TLS) handshake. (1)	8

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Question	Answer	Marks
5(a)	Two from: - Confirm/check that a copy of a file is identical to the original file (NOT source) (1) - Used to check/validate that data has been transmitted accurately (1st) – from one device to another/over a network/over the internet (1) - It is used for the whole files of data (1) - Provide a unique identifier for data (1)	2
5(b)	Three from: - Calculated by algorithms (1) - Use a hash function (1) - Adds number of bytes in a file /accept alternative of this process (1) - Checksum/number transmitted at the end of the file (1)	3

Question	Answer	Marks
7	Eight from: Simple statement of the difference between the two (1): e.g. • Direct data is a primary source, indirect data is a secondary source. • Description of how primary and secondary are collected. Advantages MAX SIX • Direct data source/direct data is reliable/trusted (1st) – because we collected it (1) • Data needs to represent the views of a specific target audience (accept other terminology) (1st) – we can make sure a cross-representation of the group is sampled/included in the sample (1) – and so avoid sample bias (1) • ONLY Specific/relevant data is collected (1) • Indirect data includes irrelevant data (do not allow opposite) (1) • Person/people collecting the data only need to collect as much/or as little (quantity) as is needed//to suit your needs (1st) – whereas with indirect data sources, the original purpose may have not collected enough//collected too much data (1) • Collected data could be sold on to other organisations (1st) – thereby helping to recoup some of the expenses/money spent (1) • Direct data is (more) up-to-date (1) Disadvantages MAX SIX • The sample size may be small (1st) – whereas indirect data sources tend to be larger sets of data (1) • Some people/groups may not be accessible by the person collecting the data (1st) – whereas with indirect data sources, would allow data from these groups to be collected/gathered (1) • With direct data collection, people may not be at home/available to be interviewed (1st) – can result in a smaller sample size (1) • Original data may not be possible to collect because of the time of year (1) • Indirect data allows historic data to be used (1) • Direct data can take a long time to collect (1) • With direct data, data needs to be collated (NOT “processed”) (1) • Collecting direct data involves the expense of travelling/photocopying/printing questionnaires//any expense to do with a direct data method (1)	8

Question	Answer	Marks
5	Six available. First mark : Describes the type of check and the data that is being checked (student ID or Student examination score). Dependant mark: Candidate states why inaccurate data could still be entered. Three matched pairs from: - Length check on student number (1st) ONE OF - won't identify if the number is wrong (1) - Won't check invalid characters//or example (1) - Won't check if value is unique (1) - Invalid character/Type check on either (1st) ONE OF - Type check on student number won't trap fewer/more than 10 digits (1) - Type check on exam mark might still allow abnormal data to be entered (1) - Won't identify if the number is wrong (1) - Won't check if value is unique (1) - Range check on either (1st) ONE OF - Range check will only check the number is in range//will not check if number incorrect (1) - Won't identify if the number is wrong (1) - Limit check on either (1st) ONE OF - Range check will only check the number is below a higher number//will not check if number incorrect (1) - Range check only checks for exceeding the maximum value/score (1) - Won't identify if the number is wrong (1) - Check digit on student number (1st) - won't identify if the number is wrong (1)	6

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Question	Answer	Marks
5(a)	THREE marks available: Max TWO - Data is collected for a specific purpose/task (1) - Provides original data/first hand data (1) Max TWO e.g. - Observation could be used, by the scientist going to the farmland and looking (1) - Questionnaires could be distributed amongst farmers/3rd party linked to scenario (1) - Interviews could be carried out with farmers/3rd party linked to scenario (1) - Data logging/sensors could be used to gather pollution data (1).	3
5(b)	THREE marks available: Max TWO - An indirect data source is when data is obtained from a third party/is second hand (1) - Means data that was collected for a particular reason but is then used for another purpose (award concept of data being used for a different purpose) (1) Max TWO - Could collect data from local government agencies (1) - Could use information presented in scientific journals (1)	3

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Question	Answer	Marks
6	FOUR from: - Verification is checking that data has been copied correctly from the data source (1) - Validation is checking that the data entered is reasonable/sensible/follows rules (1) - Verification will not pick up the fact that the original data might be invalid (1) - Verification would pick up a transposition error/most validation checks (except a check digit) would not (1) Neither validation nor verification on their own will pick up all errors//need both to cover all possible errors. (1)	4

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Question	Answer	Marks
2(a)	ONE from: - New information is of a higher quality (accept converse argument) (any awareness shown) (1) ONE from: - Up-to date information leads to better decisions (1) - Out of date information leads to poor decisions (1) - Any relevant example used to exemplify out of date information. (1)	2
2(b)	THREE from: Max TWO from: - Information needs to be relevant to the reason why/purpose it has been collected//must meet user needs (1) - Relevant information is good (1) - Irrelevant information is not useful (1) Max TWO from: - Irrelevant information leads to poor decisions/Relevant information leads to good decisions (1) - Wastes time as has to be sifted (1) - Irrelevant information can distract from the correct analysis (1) - Any relevant example used to exemplify irrelevance. (1)	3

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Question	Answer			Marks
8(a)	Field Name	Data Type	Example of Data	5
	Number_in_Stock	Integer (1)	199	
	Location	String/alphanumeric/text (1)	07.44.03	
	Reorder_Level	Integer (1)	250	
	Price	Currency (1)	\$203.50	
	Supplier_Telephone_No.	String/alphanumeric/text(1)	07700900819	

Question	Answer	Marks
8(b)	SIX from: Three matched pairs: • Number_in_Stock – Type check (1st) must be numeric (1) • Number_in_Stock – Range check (1st) between 0 and 250 (1) • Price – Limit check (1st) must be more than zero (1) • Price – Type check (1st) must be numeric/currency (1) • Location – Format check (1st) Must be three pairs of digits separated by a full stop/NN.NN.NN (1) • Location – Length check (1st) must be 8 characters (1) • Reorder_Level – Type check (1st) must be numeric (1) • Supplier_Telephone_No – Length check (1st) must be 11 characters/digits (1).	6
8(c)	FIVE from: • Any awareness of normalising data (1) • Import the (two) tables into the relational database structure (1st) as separate tables (1) • Create/use a linking table (1) • Create/use/it needs Primary Keys for both/all tables (1) • Create/use/it needs Foreign Keys. (1st) These are needed to act as the link field between the tables (only award if BOTH of the previous two answers has been given) (1) • In the linking table (1) • Link the Primary Key(s) to the Foreign Key(s) (1) • Any awareness of one-to-many relationships (1)	5

Question	Answer	Marks
9	EIGHT from: - This method gathers direct data, which is more up to date (1) Benefits max 6: - There is no significant time delay between question and answer (1) - Reporter can more easily ask extension questions based on answers (1st) that can add to the verbal answer from the interviewee (1) - People from (e.g.) all over the country (any awareness of wide location between two parties) can be interviewed (1) - Enables interviewers/reporters to contact people in restricted/remote areas (1) - No need to travel/no travel costs for reporter (1) - No time wasted/used by reporter in travelling to interview (1) - Interview can easily be recorded for later transcription/reference (1) Drawbacks max 6: - The reduction of social cues/cannot see interviewee (1st) so e.g. body language/eye contact cannot be used as a source of extra information (1) - The interviewer/reporter has no knowledge of the situation in which the interviewee is situated (1st) so less able to create a good interview ambience (1) - Telephone line/mobile signal quality can affect responses/interview (1) - Easier for interviewee to find excuse/make up excuse/find reasons to terminate the interview (1) - Can only interview one person at a time (1) - Interviewee may not pick up the phone (1) - Sample is restricted to those people who have a telephone (1) - No way to visually verify who you are interviewing (1)	8

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Question	Answer	Marks
4(a)	THREE from: • Verification is checking that data that has been/is being entered into a computer has been copied correctly (1). • Validation is checking that the data entered is reasonable/sensible/valid/follows a set of rules. (1) • Verification will not pick up the fact that the original data might be incorrect. (1) • Validation can not tell if the data has been entered incorrectly (1) • Verification would pick up a transposition error. (1) • Verification would pick up a transcription error. (1)	3
4(b)(i)	TWO from: • It is a validation method (1) • It checks that data is possible/sensible for the context (1) • It checks that data is between an upper and lower acceptable value (1) • Sensible example (1)	2
4(b)(ii)	TWO from: • It is a verification method (1) • It is used to avoid/pick up transcription errors (1) • Two versions of the data are entered and compared (1st) – If a discrepancy between the two versions is found, user will be alerted (1) • Sensible example (1)	2

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Question	Answer	Marks
5	FOUR from: • The public key is used to encrypt the data (1) • The private key is used to decrypt the data (1) • The public key is published to everyone (1) • The private key is sent separately to the receiver (1) • The private key is kept secret (1) • Anyone with a copy of the public key can encrypt information (1) • Only the private key holder can decrypt the information (1) • It is not possible to decrypt the data from the public key (1)	4

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Question	Answer	Marks
10	EIGHT Marks available: Definition of online processing: ONE from: • User is in direct connection with a central computer (1) • A real time relationship between user and seller via the internet (1) Max EIGHT: • Give a positive impact (1st) • Give an extension that is a positive implication on a customer or business of the identified justification. (1) e.g. • Speeds up the process of an individual order (1st) results in better customer satisfaction (1) Other areas of focus • Speeds up the process of a large number of orders • Allows online purchases • Allows online financial transactions • Helps to keep track of your stock levels • Continues to collect data, therefore more accurate • Reduces the time between ordering and delivery • Automated reordering • Integration with other online systems.	8

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Question	Answer	Marks
1	One mark per bullet point to a maximum of five marks. • Information is data with context (1) and meaning (1) • This could be about e.g. a student/lecturer/employee/payroll system (1) Any examples of the data presented with context or meaning (interpretation) (max 5 – each item of data may only be used once) e.g. • Chandra could be the first name. (1) • 59 could be their age. (1)	5

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Question	Answer	Marks
2	One mark per example and reasonable expansion. • First mark – any reasonable reason why a checker may miss issues when comparing two documents (1st) • Second mark – example of the error in practice (1) Answer must be in the context of comparing two documents and not be about proof reading.	2

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Question	Answer	Marks
5	One mark per valid validation check identified (max 5) and justification given to a maximum of eight marks. Max six if only one of student number or examination score used. • A format check (1st) – To ensure student numbers match the expected format/To check/while checking if student numbers starts with letter followed by 5 digits. (1) • A length check (1st) – To ensure student numbers have the correct/expected number of digit (1) • A check digit (1st) – To help prevent/detect typos/errors during entry of student numbers/As a redundancy check for student numbers (1) • A lookup check (1st) – To help verify that a student number exists in a master list// a/the student number belongs to an actual student enlisted/registered at the school. (1) • A consistency check (1st) – To ensure no duplicate/multiple records exist for the same student number against the same examination (1) • A range check (1st) – To ensure that the exam score falls within the valid/expected range of 0 to 120 (1) • A limit check (1st) – To ensure lowest exam score of 0/highest exam score of 120 (1) • A type check (1st) – To ensure that the student number field had data entered as text only (1) • A type check (1st) – To ensure that the exam score field had data entered as numeric only (1)	8

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Question	Answer	Marks
8	One mark per bullet point to a maximum of three marks. • Authenticates packets of data (1) • Encrypts data at source (1) • Decrypts data at destination (1) • Provides secure communication between two computers/across the internet (1) • IPsec is used in a virtual private networks (VPNs) (1) • Negotiation of cryptographic keys to use during a session. (1)	3

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Question	Answer	Marks
1	One mark per bullet point to a maximum of two marks. • Age//possibly too old to be of any use (1) • Relevance//may not be about recycling (1) • Accuracy//the figures/information may be incorrect (1) • Completeness// the information may be too vague (1) • Level of detail//there may be too much or too little detail (1)	2

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Question	Answer	Marks
8	Six marks from: • Customer taps card/card is read (1) • Supermarket computer contacts customer bank (1) • Checks to see if there are sufficient funds (1) • If there are sufficient funds transaction is authorised/made//money is transferred (1) • If there are insufficient funds transaction is rejected/not made//money is not transferred (1) • Amount of the transaction is deducted from customer's bank account (1) • Amount of the transaction is credited to supermarket's bank account (1) • The transaction is closed//shown as completed. (1)	6

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Question	Answer	Marks
1(a)	Four marks from: • A larger sample size (1) • A larger/large set of data to be examined (1) • It allows data to be gathered from subjects to whom the data gatherer does not have physical access (allow concept) (1) • Using less time (to collect the data)//data is readily available//already exists (1) • Data collected indirectly has already been collated/grouped (1) • Data collected indirectly has already been processed (1) • Information can be of a higher quality. (1) • Costs of (e.g.) travel//printing//employing interviewers avoided (must state the cost saved) (1)	4
1(b)	One mark per bullet point to a maximum of four marks. • The data may be out of date (1) • The data may be costly to purchase (1) • Inappropriate//irrelevant sample (who is asked//size) (1) • Irrelevant questions asked//methods used//data gathered (1) • The data may not meet the company's needs//different purpose (1) • The data may not be specific enough//lacking detail. (1)	4

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Question	Answer	Marks
11	Max FOUR marks, TWO per paired answer • Length check (1st) – Checks that the length of the data is exactly 7 characters (1) • Format check (1st) – Checks that each item of data begins with two letters followed by 5 digits (accept exemplification) (1) • Presence check (1st) – Checks that there is data present in this field (1).	4

1

Question	Answer	Marks
13	One mark from: - Definition of encryption (1) Max 6 from: <i>Benefits:</i> - Content of the web page cannot be understood by any third-party (accept any awareness) - OR - Can only be understood when decrypted//cannot be understood without decryption key (1) - Only access by authorised users (1) - Encrypted connections make man-in-the-middle attacks ineffective (1) - Reduces the impact of would be/potential hackers (1) - Dissuade/deters hackers (1) - Content of the web page (being transferred) is kept private (1) - Gives users confidence (1) Max 6 from: <i>Drawbacks:</i> - Encryption uses a lot of server resources (1) - Requires processing power/memory (1) - Encryption introduces latencies/delays//takes more time to transfer data (1) - Connections take longer to set up (1) <u>More</u> data exchanges as keys have to be exchanged (1) - Need to buy an SSL certificate (1) issued by certificate authorities (1) - There are proxy caching problems. (1)	8

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Question	Answer	Marks
1	Max 2 for each method (max two methods): Descriptions <u>must</u> match the method. • Questionnaire (1st) – A remote method of gathering data (1) – Is a fixed set of questions that cannot be changed (1) – Completed in respondent's own time/location (1) – Can be online or printed on paper//in physical form (1) • Interviews (1st) – The interviewer(s) asks the questions//questions are asked face to face (1) – Answers are gathered in real time (1) – In structured interviews the questions are asked in the same order for each interviewee and with a choice of answers (1) – Unstructured interviews can be different for each interviewee//interviewees can expand on their answers/usually no pre-set list of answers (1) – Interviewer can ask follow-up questions (1) • Data logging (1st) – Uses sensors to collect data (1) – Can be used to collect data over a period of time (1) – Either continuously or at set/regular intervals. (1) • Document analysis (1st) – Examining existing documents (1) to create new information (1)	4

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Question	Answer	Marks
2	One mark per bullet point to a maximum of five marks. • SSL stands for Secure Socket Layer and TLS stands for Transport Layer Security (1) • They provide network security (1) • TLS/SSL are used in web browsing/email/Internet faxing/instant Messaging/Voice over IP/VoIP (must have at least two examples) (1) • Client-server applications use TLS in a network to try to prevent Eavesdropping (1) • Encrypts data sent from users to website//data sent from websites to users (1) • Certificates are sent (1) and confirmed with a trusted authority (1) • They enable credit card/online payments to be made securely (1) • SSL/TLS requires a handshake to be carried out. (1) • SSL/TLS is used by the presentation layer//SSL/TLS operate/work between the application layer and the transport layer (1)	5

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Question	Answer	Marks
1(a)	One from: - Data is raw facts (1) - Data has no meaning (1) - Data is unprocessed/unorganised (1) - No context (1)	1
1(b)	One from: - Information is data with meaning (1) - Information is data in context (1)	1

Mark scheme · 9626/11 and 9626/13 Oct/Nov 2025 · Question 2

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Question	Answer	Marks
2(a)	Three from: - She will be able to provide a better quality of service (1) - She will have better quality information about her patients (1) - She can get exactly the data she wants//can ask follow-up questions//concept of relevancy (1) - Data is more reliable/trustworthy//she knows where it came from (1) - Data is more up to date (1) - Data only relates to the sample/patient that she want/collects from (1)	3
2(b)	Three from: - No organisation of data (1) - No pre-analysis of data (1) - Requires access to the source//patient is needed (1) - Prone to human error (1) - May suffer from bias (1) - May require specialised equipment (1) - No peer review (1)	3

Mark scheme · 9626/11 and 9626/13 Oct/Nov 2025 · Question 10

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Question	Answer	Marks
10	Six from: - It uses sensors and output/actuator (1st) - A valid example of input e.g. proximity sensors (1) - A valid example of output e.g. thrusters (1) - Real time processing occurs in a (micro-)processor (1) - It uses a process called feedback (1) - Output from the system affects the input/sensors/environment (1st) - Input from the sensors is used to alter the output (1) - Valid example of action e.g. trajectory correction (1) - Description of the output/input system (as a whole) in action e.g. sensor detect object in path of rocket and fires thrusters to avoid (1) - Real time processing produces an extremely/fast response(s) (to avoid critical incidents) (1) - Real time processing runs continuously (1) - Real time processing can run autonomously (1)	6

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Question	Answer	Marks
1	Three from: • Accuracy/free from error (1) • Completeness (1) • Level of detail (1) • Relevance (1) • Time/age/out of date (1)	3

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Question	Answer	Marks
2	Three from: • Used to check the file has been transmitted correctly (any awareness of verification/checking of file transfer) (1) • The control total is done only on numeric data (1) • The control total is a calculated value/any awareness of calculation to achieve a total (1) • The control total is added to the file (to be transmitted) (1) • The control total is recalculated when the file has been transmitted (must have concept of “after transmission” linked to the recalculation) (1) • Statement about outcome based on comparison of two totals (1)	3

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Question	Answer	Marks
7(a)	Four from: - Decide on what you want to learn/know (1) - Identify/prepare questions (1) - Decide on interviewer (1) - Identify population/decide which group of staff to interview (1) - Choose suitable date/time (1) - Choose suitable location (any indication of choice OR criteria) (1) - Select/invite attendees/employees (1) - Choose suitable timetable/length (1) - Prepare suitable recording methodology (1)	4
7(b)	Four from: Analysis - You can create complex reports (1st) - allows to combine data from multiple tables (1) - You can create complex queries (1st) ... (1) - You can create dynamic queries (1st) ... (1) Organisation Quicker to search (1st) - Because data is held in tables (1) Consistency - Data is not repeated/reduced redundant data/only entered once (1st) - Quicker/easier to modify the data (1) - File size can be smaller/frees up (NOT “creates”) storage space (1) - Reduced chance for errors (1)	4

Mark scheme · 9626/12 Oct/Nov 2025 · Question 9

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Question	Answer	Marks
9	Six from: - Definition of encryption (1) - Protects data/data “is safe”/improves data security (1) - Data remains protected if pen drive is lost (1) - Proves that the data has maintained its integrity (1) - Files cannot be understood unless decrypted (must have concept of encryption/decryption) (1) - Files cannot be understood by an unauthorised user/anyone without the decryption key (1) - Remains encrypted regardless of where it is moved to (1) - Reduces motivation to steal pen drive (1) - Increased confidence in saving sensitive data/files on the pen drive (1)	6

Mark scheme · 9626/12 Feb/March 2026 · Question 2

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Question	Answer	Marks
2	ONE mark available: • They do not have meaning (1) • They are not in context (1) • They have not been processed (1) • They are raw data (1)	1

Mark scheme · 9626/12 Feb/March 2026 · Question 10

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Question	Answer	Marks								
10(a)	THREE marks available: <table><tr><th>Data title</th><th>Most appropriate data type</th></tr><tr><td>employee_name</td><td>Text/alphanumeric (1)</td></tr><tr><td>hourly_rate_of_pay</td><td>Currency/numeric (1)</td></tr><tr><td>was_employee_born_in_france?</td><td>Boolean (1)</td></tr></table>	Data title	Most appropriate data type	employee_name	Text/alphanumeric (1)	hourly_rate_of_pay	Currency/numeric (1)	was_employee_born_in_france?	Boolean (1)	3
Data title	Most appropriate data type									
employee_name	Text/alphanumeric (1)									
hourly_rate_of_pay	Currency/numeric (1)									
was_employee_born_in_france?	Boolean (1)									
10(b)(i)	TWO marks available: First mark for identifying a suitable validation rule/test (1st) Second mark for a description of that test (1) • Range (1st) – to check value is BETWEEN two figures (1) • Type – to check data is in numeric/currency • Format check – to check entered as (e.g.) \$3.14//description of • Lookup check – to check against a list of values • Limit check – to check that is not below//above a certain figure • Character check – to check that is a decimal/real	2								

Question	Answer	Marks
10(b)(ii)	TWO marks available: First mark for identifying a suitable validation test (1st) Second mark for a description of that test (1) • Range – to check value is BETWEEN two figures • Type – to check data is in numeric/date • Format check – to check entered as (e.g.) dd/mm/yyyy • Limit check – to check that is not below/above a certain figure • Character check – to check that is an integer	2
10(c)	THREE marks available: • Age is based on the calculation between date of birth and today (1) • Age constantly increases with time//age changes every year (1) • A calculated field allows for a changing/dynamic figure (1) • The result shown//age is constantly accurate (1st) Reduced chance of human error resulting in the wrong number being stored (1)	3

Question	Answer	Marks	Guidance
1(a)	TWO marks available, one per complete difference: - Direct data is used for the same purpose for what it was collected.//Indirect data is used for a different purpose to what it was collected for. (1) - Direct data is collected by/for you/by the user.//Indirect data is obtained from someone else. (1) - Indirect data is from a secondary source//direct data is from a primary source (1) - Direct data is collected first hand//secondary data is second hand data (1)	2	Do Not Award - Direct data may be free, indirect data you may have to pay for - Swapping used and collected
1(b)	TWO marks available: - Candidate identifies a suitable circumstance (1st): e.g. Project needs a lot of data/named example. Project needs to be completed very soon/named example. Direct data would be too expensive to gather/named example. Direct data would be dangerous to gather/named example. Data to be gathered refers to something in the past/named example. Data may be seasonal and it's the wrong season Data is available now Direct data does not exist - Candidate explains why this circumstance would be better suited to use indirect data (MUST match circumstance) (1) e.g. Because collecting a lot of data would take time/beyond the scope of the person who needs it to collect. Because data would be gathered already. Because data is being gathered from a warzone/identified place or circumstance of danger. Because situation no longer exists for the data to be gathered.	2	

Question	Answer	Marks	Guidance
1(a)(i)	ONE from: - Questionnaire (e.g.,) where questions are asked//customers are asked their opinions (1) - Interview is (e.g.) a face to face meeting/questions are asked (1) - Observation where situations are watched (1) - Surveys (e.g.) where questions are asked (1)	1	Marks may only be awarded where the method has been stated. Answer MUST do more than identify a method
1(a)(ii)	ONE from: e.g. - Textbooks (1) - Journals (1) - Websites//research published online (1) - Someone else's (e.g.) questionnaire (1) - Market research DONE FOR someone else (1) - Government statistics (1) - Newspapers/magazines (1)	1	Do Not Award: - Weather data - Census data - Electoral register - Any source that would not include expenditure patterns - Any source where it is not clearly someone else's

Question	Answer	Marks	Guidance
1(b)	Four marks available: TWO from relevance: • Reduces the chance of excess/unnecessary data (accept converse) (1) • The businesswoman has control over what information is gathered (accept exemplification) e.g. ask questions about sports goods (1) • The businesswoman can design the method/stated method, so can ensure that the (e.g.) questions asked gather the information she needs (1) • Can ensure data is gathered from the town where she will open the shop (1) • Data is obtained/collected with the set purpose in mind (1) TWO from age: • The questions are being asked now (1) • The age of the data is a measure of how long ago it was collected (1) • Use of a direct data source means that data can be/is more up-to-date (1) • Patterns of behaviour/attitudes change over time//what was correct (e.g.) twenty years ago is not necessarily correct now (accept converse about direct data) (NOT 'things could have changed' – TV (1)	4	

Question	Answer	Marks	Guidance
1(c)	Four marks available. Two per matched explanation: Identify reason (1) and state why it is a reason/why beneficial (1) Areas for consideration (Note: may also be used as expansion. Mark as candidate's intension) • Larger sample size (must be the concept of more data/more people asked) • Less biased/more accurate/more reliable NB accuracy or reliability could be used as an extension of (e.g.) less biased for two marks • Wider variation/type of people • Seasonal nature of some sports • Convenience • No time to collect data so use someone else's • Avoidance of a cost (1) such as (1) • Quicker to achieve results (of analysis) Exemplar answers • The sample size may be larger (1) therefore the results gathered are less likely to be biased (1) • She may not have the time to collect the data (1) and so uses data gathered by someone else (1)	4	

Question	Answer	Marks	Guidance
4(a)	FOUR from: • Both methods check that the information entered matches the <u>original/the book/the source</u>//has been inputted correctly (1) Accuracy • Neither method checks if the source data is correct (1) • Double data entry is more accurate (accept converse) (1) • Visual checking may be affected by human error//checker may miss error//double data entry does not have error (when checked) (1) Who does it • Visual checking is carried out by a person/user/manually//For double data entry, it is done by the computer (1) • For double data entry, the user is only involved if an issue is found (1) Speed • Double data entry is a slower process (accept converse) (1) General • For visual checking, the data is only entered once//for double data entry the data must be entered twice (1) • For double data entry, a second copy is temporarily created (1)	4	

Question	Answer	Marks	Guidance
4(b)	THREE marks available for ONE identified method and suitable description - Parity check (1st) Counts the numbers of 1s in (each byte) of data and adds a 1 if it is an uneven number/adds a zero if it is an even number (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1) - Checksum (1st) Uses a hash function (1) Counts the number of bytes/bits in a file//takes file size and produces a number (1) Concept of recalculating and checking again (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1) - Hash total//hash sum (1st) A number is calculated from the content of each field of data. (1) Concept of recalculating and checking again (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1) - Control total (1st) A number is calculated from the fields holding numerical data. (1) Concept of recalculating and checking again (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1)	3	Do not accept: - Visual checking - Double data entry

Question	Answer	Marks	Guidance
5(a)	FOUR from: STRUCTURE - In a flat-file database, there will be one table// in a relational database there will be more than one table (1) ACCURACY (accept converse for each) - A relational database is less likely to have data entry errors (1) (For the same data,) a relational database will be smaller//save storage space (1) THE DATA - A relational database will have less duplicated data/redundant data (1) OR - a flat-file database can have the same data repeated (many times)//have lots of redundancy (1) - Each record in a relational database only has to be entered once//in a flat-file database, the same data must be entered every time it is needed (1) - Any mention of data integrity being applied to a relational database (1) TIME (accept converse for each) - Relational databases take less time to maintain/edit the data (1) - Relational database takes longer to edit the structure (1) - Querying a relational database is faster than querying a flat-file databases (1) COMPLEXITY (do not accept converse) - Relational database is more complex (1) CAN ALSO AWARD IF STATED SEPARATELY - Relational database can create complex queries (1) - Relational database can create complex reports (1)	4	Do Not accept - Less time/more time to plan/make - Easier to create/update - Any reference to files rather than tables

Question	Answer	Marks	Guidance
5(b)	Three marks available: Structure: - Identifies the validation type (1st) - Describes a rule (1) - States why it is important//why is it used//what error does it stop (1) e.g. - Presence check (1st) Each field has to contain an item of data//cannot be left empty (1) To check that data has been entered into the field//because the data is necessary (1)	3	

Question	Answer	Marks	Guidance
8	FOUR from: Batch processing • All processing is done at the same time//processing is a single process//all at once (1) • Done at a fixed time (1) • Likely to be at the end of the day/when the supermarket has closed (1) • No action occurs until sometime after the sale (1) Real time processing • Real-time processing is done immediately (1) • As soon as an item is sold (1) • Supermarkets can order immediately stock levels become low//allows just in time stock control (1) General • Information for real-time processing is (more) up to date (1) • If the supermarket does not sell out before the close each day, both methods are equally effective (1) • If the supermarket sells out before the close each day, real-time processing allows items to be ordered immediately (and restocked that day) (1) • Errors are not identified immediately with batch processing (accept converse) (1) • Data may be lost (between sale and processing) for batch processing//data is less likely to be lost (1)	4	Accept converse arguments for the general section ONLY

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Question	Answer	Marks	Guidance
1	THREE marks available: TWO from: - Data consists of raw facts and figures which require processing for interpretation (1) - Data does not have meaning (1) - Information is data that has a context (1) Allow converse for the above ONE mark for: - Use of ONE example to describe BOTH data and information (clearly stated as such) (1)	3	

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Question	Answer	Marks	Guidance
3	FOUR from: Identify a benefit and state why it is a benefit OR cause of benefit Benefit e.g. - Data is safer/protected/secure (1st) - Stolen data cannot be understood (1st) - Reduced desire to hack (1st) - Data is unusable if stolen/accessed (1st) Explanation e.g. Any suitable explanation	4	

4(a)	TWO marks available: • High quality data is (more) accurate/reliable/detailed/relevant/up-to-date/complete//data needs to be/should be accurate (1) • and decisions based on them are more likely to be correct//decreases risk of making an incorrect decision (1) • provides better insights into shopping habits (1) Answer may be focused on one factor: e.g. • Data must be up-to-date (1) because trends change over time /you know what is popular now (1) ACCEPT CONVERSE of these arguments where applicable	2	
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Question	Answer	Marks	Guidance
4(b)	EIGHT from: - The report holds information that may be useful, but there are reasons why it may not be (1) Three marks may be awarded where relevant issues are identified, but not explained, where no other marks would be awarded overall other than Mark Point 1 (above). Benefits (Max six) e.g. - It is data that has been collected for the similar purpose (1) - The data already exists (1) so there is no time required to gather the data (1) decision can be made more quickly (1) - The data has already been organised (1) therefore it is not raw data, so no organisation required (1) - Data will be complete/lots of detail (1) - It is from a larger sample (1) removes bias (1) - Allows access to data that may be (currently) impossible/difficult to gather (1) COSTS - The costs of gathering the data (to the shop owner) are zero/minimal/reduced (1) - Identified saving cost of gathering data (1)	8	Answers must be linked to using the data to identify shopping habits. Do Not Award - Data is untrustworthy - Data is up-to-date

Question	Answer	Marks	Guidance
4(b)	Drawbacks (Max six) • The data sample is national (1st) therefore not focussed enough//irrelevant (1) • The data is old// is now out of date (1) so trends//shopping habits may have changed (1) • Report may be biased (NOT 'false') (1) Conclusion One mark for a value judgement/conclusion based on two arguments (positive or negative or both) (1)		